

Space Shooter - Game Design Document

1. Game Overview

Core Concept

A top-down space shooter where the player controls a spaceship, destroys asteroids and enemy ships, collects power-ups, and survives as long as possible while achieving the highest score.

Target Platform

- Web browser (HTML5/Canvas or WebGL)
- Mobile-friendly controls optional
- Single-player experience

Game Genre

- Arcade shooter
- Survival/Score attack
- Action

2. Gameplay Mechanics

Player Controls

- **Movement:** Arrow keys or WASD for 8-directional movement
- **Shooting:** Spacebar or mouse click for continuous firing
- **Special:** Optional special ability key (e.g., shield activation)

Core Loop

1. Player spawns in center of screen
2. Asteroids and enemies spawn from screen edges
3. Player navigates and shoots to destroy threats
4. Destroyed objects drop power-ups and points
5. Difficulty increases over time
6. Game ends when player loses all lives

Scoring System

- **Small Asteroid:** 10 points
- **Medium Asteroid:** 25 points
- **Large Asteroid:** 50 points

- **Basic Enemy:** 100 points
- **Advanced Enemy:** 250 points
- **Bonus for collecting power-ups:** 50 points
- **Survival time bonus:** 1 point per second

3. Game Objects

Player Ship

- **Health:** 3 lives/hits
- **Speed:** Moderate, responsive movement
- **Weapon:** Basic laser that can be upgraded
- **Size:** Small hitbox for precise dodging
- **Visual:** Classic spaceship sprite with thrust animation

Asteroids

- **Large Asteroids:**
 - Move slowly in straight lines
 - Split into 2 medium asteroids when destroyed
 - Highest point value
- **Medium Asteroids:**
 - Move at moderate speed
 - Split into 3 small asteroids when destroyed
 - Medium point value
- **Small Asteroids:**
 - Move quickly
 - Completely destroyed when shot
 - Lowest point value but most numerous

Enemy Ships

- **Basic Fighter:**
 - Moves toward player
 - Shoots simple projectiles
 - Dies in 1 hit
- **Heavy Cruiser:**
 - Slower movement

- More health (3 hits)
- Shoots spread of bullets
- Spawns less frequently

Power-ups

- **Rapid Fire:** Increases shooting speed for 10 seconds
- **Multi-Shot:** Shoots 3 bullets simultaneously for 15 seconds
- **Shield:** Temporary invincibility for 5 seconds
- **Extra Life:** Adds one life to player
- **Score Multiplier:** 2x points for 20 seconds

Projectiles

- **Player Bullets:** Fast, small, yellow/blue color
- **Enemy Bullets:** Slower, red color, destroys on contact with player

4. Game Progression

Difficulty Scaling

- **Wave 1-5:** Only asteroids, slow spawn rate
- **Wave 6-10:** Introduction of basic enemies
- **Wave 11-15:** Increased spawn rate, medium asteroids
- **Wave 16-20:** Heavy cruisers appear
- **Wave 21+:** Maximum difficulty, all enemy types

Spawn Patterns

- **Early Game:** 1-2 large asteroids per wave
- **Mid Game:** Mix of asteroids and 1-2 basic enemies
- **Late Game:** Multiple enemies with asteroid fields

Power-up Frequency

- Power-ups have 15% chance to drop from destroyed enemies
- 5% chance from destroyed asteroids
- Only one power-up active at a time (new ones override old)

5. Technical Requirements

Performance Targets

- 60 FPS on modern browsers
- Smooth movement and collision detection
- Efficient particle systems for explosions
- Memory management for bullet cleanup

Screen Resolution

- Primary: 1280x720 (16:9 aspect ratio)
- Scalable UI elements
- Responsive design for different screen sizes

Audio Requirements

- Background music loop (space/electronic theme)
- Sound effects: shooting, explosions, power-up pickup, enemy hit
- Volume controls in settings

6. User Interface

HUD Elements

- **Score:** Top-left corner
- **Lives Remaining:** Top-right corner (heart icons)
- **Current Wave:** Top-center
- **Active Power-up:** Bottom-right with timer
- **High Score:** Displayed after game over

Menus

- **Main Menu:** Play, Instructions, Settings
- **Pause Menu:** Resume, Restart, Main Menu
- **Game Over Screen:** Final Score, High Score, Play Again
- **Instructions:** Control scheme and gameplay tips

Visual Style

- Dark space background with stars
- Bright, contrasting colors for game objects

- Particle effects for explosions and thrust
- Clean, readable font for UI text

7. Game States

State Management

1. **Loading:** Asset loading screen
2. **Main Menu:** Initial player entry point
3. **Playing:** Active gameplay state
4. **Paused:** Temporary stop with overlay
5. **Game Over:** Score display and restart options
6. **Settings:** Audio and control configuration

8. Asset Requirements

Graphics Needed

- Player ship sprite (with animation frames)
- 3 asteroid sizes (with rotation frames)
- 2 enemy ship types
- Bullet sprites (player and enemy)
- Power-up icons (5 types)
- Explosion animation frames
- Background starfield
- UI elements and fonts

Audio Needed

- Shooting sound (short, punchy)
- Explosion sounds (3 variations)
- Power-up pickup sound
- Enemy hit sound
- Background music (looping)
- Game over sound

Estimated Development Time

- **Basic Gameplay:** 2-3 weeks
- **Polish and Effects:** 1-2 weeks

- **Audio Integration:** 3-5 days
- **Testing and Balance:** 1 week
- **Total:** 4-6 weeks (part-time development)

9. Success Metrics

Player Engagement

- Average session length: 3-5 minutes
- High score progression over multiple plays
- Intuitive controls requiring no tutorial

Technical Performance

- Consistent frame rate across devices
- Quick load times (<3 seconds)
- Responsive input with minimal lag

This design provides a solid foundation for a classic arcade-style space shooter that's both fun to play and achievable to develop.